Meehir Prabhakar

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Education

VIT-AP University India
Bachelor of Technology - BTech, Computer Science and Engineering *Aug. 2023 – Expected Jun 2027*

Experience

TechtoGreen Drone & Robotics Private Limited Amaravati, Andhra Pradesh, India
AI/ML and Computer Vision Intern *Dec. 2024 – May 2026*

- **Autonomous Vision Systems:** Engineered a multi-model vision pipeline (YOLO, VGGT, OpenCV) for real-time crop pathology detection and worker safety monitoring.
- **Depth & Vegetation Analysis:** Developed a dual-model vision pipeline combining VGGT monocular depth estimation and YOLO instance segmentation. Engineered a bitwise mask intersection algorithm to filter foliage within a 1.0-meter proximity threshold.
- **SLAM & Spatial Computing:** Developed binocular visual SLAM and odometry architectures using StereoSGBM disparity mapping, ORB/FAST feature extraction, and Levenberg-Marquardt pose optimization.
- **Algorithmic Optimization:** Designed custom Non-Maximum Suppression (NMS) algorithms, implemented ArcFace embeddings with FAISS for asynchronous POI tracking, and trained Isolation Forests for anomaly detection.
- **Hardware Integration & Backend:** Synchronized parallel inference with MAVLink telemetry and YDLIDAR G2 point clouds via C++. Built a FastAPI backend to generate PDF agronomic reports.

The Machine Learning CLUB VITAP Amaravati, Andhra Pradesh, India
Technical & Project Team Member *Oct. 2023 – May 2025*

- Engineered a 3D survival game in Godot (GDScript) featuring state-persistent UIs, dynamic minimaps, and NavigationMesh-based enemy AI pathfinding with progressive difficulty scaling.
- Hosted on GitHub: github.com/MLC-VIT-AP/PAC_3D

Projects

AI SIEM & MDR Pipeline | *Python, Scikit-Learn, SHAP, LIME* Jan. 2026 – May 2026

- Engineered an AI-driven SIEM/MDR framework that normalizes heterogeneous telemetry (network flows, web apps, OS logs) into a unified JSON schema.
- Deployed Isolation Forests for L3/L4 network anomaly detection and One-Class SVMs with TF-IDF N-grams for L7 web/auth threat isolation.
- Built a central Fusion Engine utilizing a temporal sliding window for cross-layer cyber threat correlation, integrating SHAP and LIME for explainability.
- Hosted on GitHub: github.com/Meehir5907/Chakshu

Autonomous Sewage Maintenance Submarine | *Python, YOLOv8n, NCNN, Open3D, RPi 5* Jul. 2024 – Nov. 2025

- Engineered a neutrally buoyant autonomous submarine for deep sewage inspection, utilizing an isolated 11.1V dual-battery architecture to prevent CPU brownouts.
- Deployed an NCNN-optimized YOLOv8n instance segmentation model on a headless Raspberry Pi 5 to identify and map structural cracks in real-time at 3.0 FPS.
- Designed a mechanical SONAR imaging system (DYP-A08) using Open3D noise reduction and Matplotlib bicubic interpolation to generate depth heatmaps for zero-visibility navigation.
- Programmed 3D propulsion and arm actuation via `lgpio`, controlling RS2205 brushless motors via 30A ESCs and a PCA9685 PWM controller.

Awards & Achievements

1st Place (Dept. of AI/DS), Innovative Model: Secured 1st place at Siddhartha Academy of Higher Education Innovation Day (Oct. 2025) in department of Artificial Intelligence / Data Science.
AI Hackathon 2025: AIMERS Letter of Appreciation (Feb. 2025).
Ideathon 2K25: Innovation Showcase Participant at Chalapathi Institute of Engineering & Technology (Mar. 2025).

Technical Skills

Languages: Python, C, C++, SQL, Go, Bash, Zsh, Lua, Kotlin, GDScript, Scala
AI, ML & Vision: PyTorch, TensorFlow, Keras, OpenCV, YOLO, MediaPipe, FAISS, Transformers, SHAP, LIME
Systems & Tools: Linux (KVM, Kernel Tuning, Cron), Docker, Kubernetes, Git/GitHub, FastAPI, QML, Volatility 3, AVML, Wireshark, Snort, Suricata
Hardware & Edge: ESP32, Arduino, LiDAR (YDLIDAR), MAVLink Telemetry, Raspberry Pi
Core Domains: Computer Vision, Visual SLAM, Cybersecurity (SIEM/MDR) Pipelines, Digital & Memory Forensics, Unsupervised Learning